

# Quantitative Research Methods

## Chapter 13: Multiple Testing

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HSBI

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# The course at a glance

| Lecture | Topic                                                          |
|---------|----------------------------------------------------------------|
| 1       | Ch. 1 + 2 — Introduction; what is statistical learning         |
| 2       | Ch. 2 — Accuracy, bias–variance, Bayes classifier, KNN         |
| 3–4     | Ch. 3 — Linear regression (estimation, inference, diagnostics) |
| 5–6     | Ch. 4 — Classification (logistic, LDA/QDA, naive Bayes, ROC)   |
| 7       | Ch. 5 — Resampling (validation, CV, bootstrap)                 |
| 8       | Ch. 6 — Model selection and regularisation (ridge, lasso)      |
| 9       | Ch. 7 — Moving beyond linearity (splines, GAMs)                |
| 10      | Ch. 8 — Tree-based methods (trees, forests, boosting)          |
| 11      | Ch. 10 — Deep learning (MLPs, CNNs, training)                  |
| ► 12    | <b>Ch. 13 — Multiple testing (FWER, FDR)</b>                   |

Twelve sessions of 180 minutes. ► marks today's chapter. Short exercises every ~20 min; extended exercises every ~45 min; each chapter has a companion lab notebook.

# Contents

- 1 The Problem
- 2 FWER
- 3 FDR
- 4 Practice
- 5 Python Lab
- 6 Summary

Optional and advanced material is collected in the **appendix** at the end of the deck.

# Notation in this chapter

| Symbol         | Meaning                                                                                    |
|----------------|--------------------------------------------------------------------------------------------|
| $m$            | number of hypotheses tested simultaneously                                                 |
| $\alpha$       | per-test significance level; target FWER                                                   |
| $H_{0j}$       | $j$ -th null hypothesis in the family                                                      |
| $p_j, p_{(j)}$ | $p$ -value of test $j$ ; the $j$ -th smallest (sorted) $p$ -value                          |
| $V$            | number of false positives (true nulls rejected)                                            |
| $R$            | total number of rejections, $R = V + S$                                                    |
| $S$            | number of true positives (false nulls rejected)                                            |
| FWER           | family-wise error rate $\Pr(V \geq 1)$                                                     |
| FDR            | false discovery rate $E[V / \max(R, 1)]$                                                   |
| $q$            | target FDR level of Benjamini–Hochberg                                                     |
| Power          | $E[S] / m_1$ , with $m_0$ and $m_1$ the number of true and false nulls ( $m_0 + m_1 = m$ ) |

## Sources and references

### Primary textbook

James et al. (2023). *An Introduction to Statistical Learning, with Applications in Python*. Springer.

# Learning objectives

By the end of this chapter you will be able to:

- **Explain** the multiple-testing problem and its consequences.
- **Apply** Bonferroni and Holm corrections to control family-wise error rate (FWER).
- **Apply** the Benjamini–Hochberg procedure to control false discovery rate (FDR).
- **Use** resampling-based  $p$ -values when the null distribution is hard to derive.
- **Avoid** the post-selection-inference and  $p$ -hacking pitfalls.

# Roadmap of this chapter

## Honest inference at scale

- 1 The multiple-testing problem.
- 2 FWER controls: Bonferroni, Holm.
- 3 FDR control: Benjamini–Hochberg.
- 4 Resampling-based  $p$ -values.
- 5 Practical issues:  $p$ -hacking, post-selection inference.

# Outline

- 1 The Problem
- 2 FWER
- 3 FDR
- 4 Practice
- 5 Python Lab
- 6 Summary



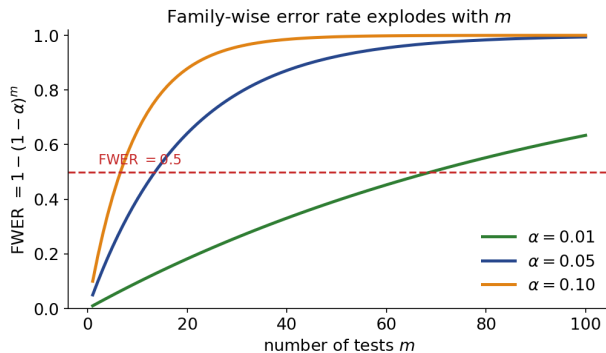
# Why naive testing fails at scale

Test  $m$  hypotheses simultaneously, each at level  $\alpha = 0.05$ .

## The arithmetic

- Even if all  $m$  nulls are true, we expect  $0.05m$  false rejections.
- Test 10 000 genes  $\Rightarrow \sim 500$  false discoveries by chance.
- Naive thresholding fails dramatically.

# Why the family-wise error rate explodes



## Takeaway

With  $\alpha = 0.05$  the chance of  $\geq 1$  false positive passes 50% by only  $m = 14$  tests and approaches certainty as  $m$  grows.

# The classical hypothesis-test refresher

## How to read this

- $H_0$ : null hypothesis.  $H_1$ : alternative.
- Type I error: reject  $H_0$  when true.
- Type II error: fail to reject when false.
- $p$ -value: probability under  $H_0$  of a test statistic at least as extreme as observed.
- “Reject  $H_0$  if  $p < \alpha$ ” controls the per-test Type I error rate at  $\alpha$ .

## Takeaway

A  $p$ -value is computed *assuming*  $H_0$  is true; it is **not** the probability that  $H_0$  is true, and a small  $p$  is not proof of a large effect.

# The decision-table view

|             | Not rejected | Rejected | Total |
|-------------|--------------|----------|-------|
| $H_0$ true  | $U$          | $V$      | $m_0$ |
| $H_0$ false | $T$          | $S$      | $m_1$ |
| Total       | $m - R$      | $R$      | $m$   |

## How to read this

- Family-wise error rate (FWER):  $\Pr(V \geq 1)$ .
- False discovery rate (FDR):  $E[V/R \mid R > 0] \Pr(R > 0)$ .
- Power:  $E[S]/m_1$ .

## Exercise 13.1 — The multiple-testing problem [Concept]

### Exercise

- 1 Define a Type I error and a Type II error in terms of  $H_0$ .
- 2 Suppose  $m$  hypotheses are tested independently, each at level  $\alpha = 0.05$ , and *all* nulls are in fact true. Write the probability of making at least one false rejection (the FWER) as a function of  $m$ .
- 3 Evaluate this probability for  $m = 10$ .
- 4 Find the smallest  $m$  for which the FWER reaches 0.5.

# Solution — Exercise 13.1 (1/2)

## Solution

**Method.** The FWER is easiest to reach through its complement: rather than count “at least one false positive” directly, find the single event “no false positive at all” and subtract from 1.

### Part 1 — error types.

- Type I (false positive): rejecting  $H_0$  when it is true.
- Type II (false negative): failing to reject  $H_0$  when it is false.

**Part 2 — FWER formula.** One test avoids a false positive with probability  $1 - \alpha$ . Under independence the  $m$  tests avoid it *jointly* with probability  $(1 - \alpha)^m$  (independent probabilities multiply), so the complement is

$$\text{FWER} = \Pr(\text{at least one false positive}) = 1 - (1 - \alpha)^m.$$

*Common mistake: adding the per-test rates to get  $m\alpha$  — that only approximates the FWER (the first-order term) and can exceed 1.*

## Solution — Exercise 13.1 (2/2)

## Solution

**Part 3** —  $m = 10$ .

$$1 - (0.95)^{10} = 1 - 0.5987 = 0.4013 \approx 40\%.$$

A 5% per-test rate has ballooned to a 40% chance of at least one spurious “discovery”. **Part 4** — FWER = 0.5. Solve  $1 - 0.95^m \geq 0.5$ , i.e.  $0.95^m \leq 0.5$ :

$$m \geq \frac{\ln 0.5}{\ln 0.95} = \frac{-0.6931}{-0.05129} = 13.51,$$

so  $m = 14$  tests already give a better-than-even chance of a false positive. This is exactly why corrections are needed.

*Common mistake: dividing by  $\ln 0.95$  without flipping the inequality. Since  $\ln 0.95 < 0$ ,  $0.95^m \leq 0.5 \iff m \ln 0.95 \geq \ln 0.5 \iff m \geq \ln 0.5 / \ln 0.95$  — the sign flips at the last step.*

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# Bonferroni: split the error budget equally

## Rule

Reject  $H_{0j}$  if  $p_j \leq \alpha/m$ .

## How to read this

- $m$ : number of tests in the family;  $\alpha$ : target FWER (e.g. 0.05).
- Divide the budget evenly: each test is judged at the stricter level  $\alpha/m$ .
- Equivalently, adjust each  $p$ -value to  $\tilde{p}_j = \min(1, m p_j)$  and compare to  $\alpha$ .

## Takeaway

Controls FWER at  $\alpha$  under *any* dependence. Conservative: it throws away power, but is simple, popular, and a good starting point.

# Holm's step-down procedure (uniformly beats Bonferroni)

## Algorithm

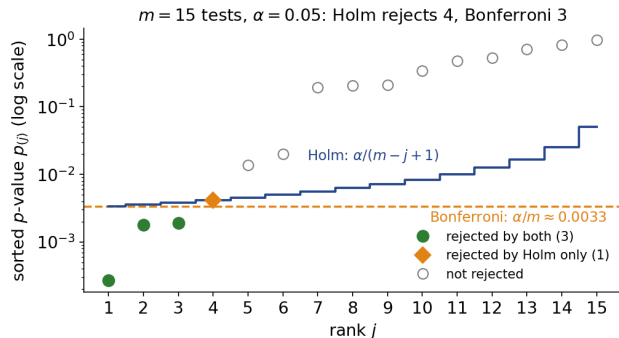
- 1 Order  $p$ -values  $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$ .
- 2 Find the smallest  $j$  with  $p_{(j)} > \alpha / (m - j + 1)$ .
- 3 Reject all  $H_{0(k)}$  for  $k < j$ .

## Why it is more powerful

- The thresholds *grow* down the list:  $\frac{\alpha}{m}, \frac{\alpha}{m-1}, \dots, \alpha$  — the first equals Bonferroni, every later one is looser.
- “Step-down”: start at the smallest  $p$ -value and walk down, stopping at the first test that fails; keep all larger  $p$ -values.

## Takeaway

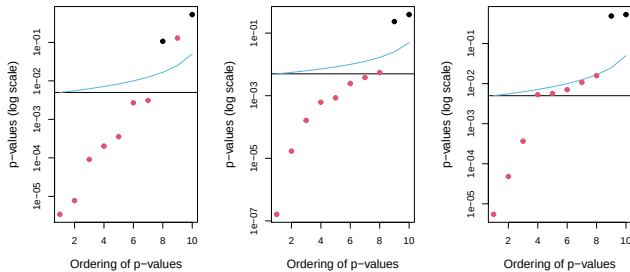
Strictly more powerful than Bonferroni, yet still controls FWER at  $\alpha$ .

Bonferroni vs. Holm on 15 simulated  $p$ -values

## Takeaway

Bonferroni's flat bar  $\alpha/m \approx 0.0033$  rejects 3 hypotheses; Holm's rising thresholds  $\alpha/(m-j+1)$  additionally catch  $p_{(4)} = 0.0041$ , rejecting 4 — Holm always rejects at least as many as Bonferroni.

# Power loss with FWER control



## Takeaway

Even Holm's procedure loses substantial power as  $m$  grows: guarding against *any* false positive forces tiny thresholds, so genuine effects slip through.

Source: Figure 13.3 — ISLP, James et al. (2023).

## Exercise 13.2 — Bonferroni correction [Math]

### Exercise

You run  $m = 6$  tests and obtain the sorted  $p$ -values

$$p = (0.001, 0.008, 0.012, 0.030, 0.040, 0.600),$$

and wish to control the FWER at  $\alpha = 0.05$ .

- 1 State the Bonferroni rejection threshold.
- 2 Which hypotheses are rejected?
- 3 Compute the Bonferroni-adjusted  $p$ -values  $\tilde{p}_j = \min(1, m p_j)$  and confirm they give the same decisions.

## Solution — Exercise 13.2

## Solution

**Step 1 — threshold.**  $\alpha/m = 0.05/6 = 0.00833$ .

**Step 2 — reject if  $p_j \leq 0.00833$ .**

| $p_j$           | 0.001 | 0.008 | 0.012 | 0.030 | 0.040 | 0.600 |
|-----------------|-------|-------|-------|-------|-------|-------|
| $\leq 0.00833?$ | ✓     | ✓     | ×     | ×     | ×     | ×     |

Reject  $H_1$  and  $H_2$  only ( $0.008 < 0.00833$ ;  $0.012 > 0.00833$ ).

**Step 3 — adjusted  $p$ -values  $\tilde{p}_j = \min(1, 6p_j)$ :**

(0.006, 0.048, 0.072, 0.180, 0.240, 1.000).

Rejecting when  $\tilde{p}_j \leq 0.05$  again flags exactly  $H_1$  (0.006) and  $H_2$  (0.048). Two rejections.

**Take-away.** Bonferroni is simple and valid under any dependence, but conservative — here it misses  $H_3$  at  $p = 0.012$ .

*Common mistake:* correcting twice — compare raw  $p_j$  with  $\alpha/m$ , or adjusted  $\tilde{p}_j$  with  $\alpha$ . Comparing  $\tilde{p}_j$  with  $\alpha/m$  double-penalises.

## Exercise 13.3 — Holm's step-down procedure [Math]

### Exercise

Using the same data as Exercise 13.2,

$$p = (0.001, 0.008, 0.012, 0.030, 0.040, 0.600), \quad m = 6, \quad \alpha = 0.05,$$

apply Holm's step-down procedure. For each rank  $j$  compare  $p_{(j)}$  with  $\alpha/(m-j+1)$ , stop at the first failure, and report the rejections. Compare with Bonferroni.

## Solution — Exercise 13.3

## Solution

The  $p$ -values are already sorted. Compare each with its Holm threshold  $\alpha/(m - j + 1)$ :

| $j$ | $p_{(j)}$ | $\alpha/(m - j + 1)$ | decision               |
|-----|-----------|----------------------|------------------------|
| 1   | 0.001     | $0.05/6 = 0.00833$   | $0.001 \leq 0.00833$ ✓ |
| 2   | 0.008     | $0.05/5 = 0.01000$   | $0.008 \leq 0.01000$ ✓ |
| 3   | 0.012     | $0.05/4 = 0.01250$   | $0.012 \leq 0.01250$ ✓ |
| 4   | 0.030     | $0.05/3 = 0.01667$   | $0.030 > 0.01667$ ✗    |

The first failure is at  $j = 4$ . Holm rejects all hypotheses with rank  $< 4$ , i.e.  $H_{(1)}, H_{(2)}, H_{(3)}$  — **three** rejections.

**Comparison.** Holm also rejects  $H_{(3)}$  at  $p = 0.012$ , which Bonferroni missed. Holm is uniformly more powerful than Bonferroni while still controlling the FWER at  $\alpha$ , because its first threshold matches Bonferroni's but every later one is larger.

*Common mistake:* continuing down the list after the first failure. Holm is *step-down*: once  $p_{(4)}$  fails, ranks 4–6 are all retained — even if a later  $p_{(j)}$  were to sit below its own threshold.



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# The Benjamini–Hochberg (BH) procedure

## Algorithm (control FDR at level $q$ )

- 1 Order the  $p$ -values:  $p_{(1)} \leq \dots \leq p_{(m)}$ .
- 2 Find the largest  $L$  such that  $p_{(L)} \leq Lq/m$ .
- 3 Reject the  $L$  smallest hypotheses.

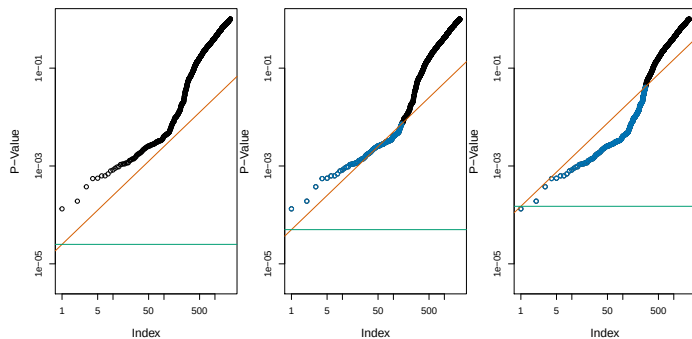
## Reading the threshold $Lq/m$

- $q$ : target FDR (e.g. 0.10);  $m$ : number of tests;  $L$ : rank in the sorted list.
- The bar  $Lq/m$  *rises* with rank, unlike Bonferroni's flat  $\alpha/m$ .
- "Step-up": scan from the *largest*  $p$ -value down; reject everything up to the first one that clears its bar.

## Takeaway

Under independence (or positive dependence), BH controls FDR at  $q$ .

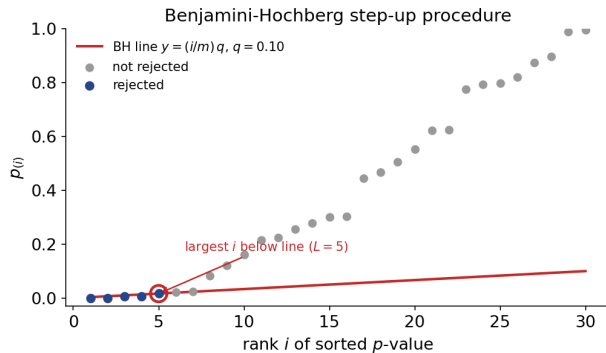
# Visualising BH



Sorted  $p$ -values vs. the line  $Lq/m$ . Reject everything below the highest crossing.

Source: Figure 13.6 — ISLP, James et al. (2023).

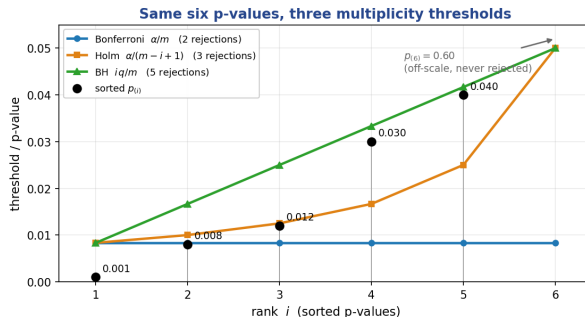
# Benjamini–Hochberg as a staircase



## Takeaway

Compare each sorted  $p_{(i)}$  with the line  $(i/m)q$ ; take the *largest* rank below it and reject everything up to there (5 of 30 here at  $q = 0.10$ ).

# Three thresholds on the same six p-values



## Takeaway

A  $p$ -value is a candidate for rejection when it sits *below* its method's curve. All three meet at rank 1; then Bonferroni stays flat while Holm and BH rise — so here they reject 2, 3 and 5 hypotheses (Exercises 13.2/13.3/13.5).

# FDR vs. FWER: which to use?

## How to read this

- Use **FWER** when even one false discovery is unacceptable (drug-approval trials).
- Use **FDR** when an exploratory list of candidates is acceptable and false positives are tolerable in moderation (genomics, A/B testing).
- BH is much more powerful than Bonferroni when  $m$  is large and a non-trivial fraction of nulls are false.

## Exercise 13.4 — FDR vs. FWER [Concept]

### Exercise

- 1 Give the definitions of the FWER and the FDR in terms of the decision-table quantities  $V$  (false rejections) and  $R$  (total rejections).
- 2 Explain why controlling the FDR is generally less conservative (more powerful) than controlling the FWER.
- 3 For each scenario, state whether FWER or FDR control is more appropriate and why: (i) a confirmatory phase-III drug trial; (ii) screening 20 000 genes for follow-up experiments.

## Solution — Exercise 13.4 (1/2)

## Solution

## Part 1 — definitions.

$$\text{FWER} = \Pr(V \geq 1), \quad \text{FDR} = E \left[ \frac{V}{R} \mid R > 0 \right] \Pr(R > 0).$$

FWER is the probability of *any* false positive; FDR is the *expected proportion* of false positives among the rejections.

**Part 2 — why FDR is more powerful.** FWER guards against a single false positive anywhere, so its threshold must shrink with  $m$  (roughly  $\alpha/m$ ). FDR only requires the false positives to be a small *fraction* of discoveries, so it tolerates a few errors when there are many true discoveries. This lets it use a larger, adaptive threshold and reject far more when many nulls are false.



# Solution — Exercise 13.4 (2/2)

## Solution

### Part 3 — scenarios.

- (i) **FWER** (Bonferroni/Holm): a single false approval is unacceptable; the cost of any Type I error is very high.
- (ii) **FDR** (Benjamini–Hochberg): an exploratory candidate list is the goal; a modest fraction of false leads is acceptable and will be filtered by follow-up experiments. FWER here would be far too conservative and kill power.

*Common mistake:* defaulting to FWER “to be safe”. Safety is not free — the stricter criterion buys its guarantee with missed true effects, so the choice is a design decision, not a formality.

## Exercise 13.5 — Benjamini–Hochberg procedure [Math]

### Exercise

Using the same six  $p$ -values,

$$p = (0.001, 0.008, 0.012, 0.030, 0.040, 0.600), \quad m = 6,$$

apply the Benjamini–Hochberg (BH) procedure at FDR level  $q = 0.05$ .

- 1 For each rank  $L$  compute the BH threshold  $Lq/m$ .
- 2 Find the largest  $L$  with  $p_{(L)} \leq Lq/m$ .
- 3 State the rejections and compare with Bonferroni and Holm.

## Solution — Exercise 13.5 (1/2)

## Solution

Steps 1–2 — compare each  $p_{(L)}$  with  $Lq/m$ , where  $q/m = 0.05/6 = 0.00833$ :

| $L$ | $p_{(L)}$ | $Lq/m$  | $p_{(L)} \leq Lq/m?$ |
|-----|-----------|---------|----------------------|
| 1   | 0.001     | 0.00833 | ✓                    |
| 2   | 0.008     | 0.01667 | ✓                    |
| 3   | 0.012     | 0.02500 | ✓                    |
| 4   | 0.030     | 0.03333 | ✓                    |
| 5   | 0.040     | 0.04167 | ✓                    |
| 6   | 0.600     | 0.05000 | ×                    |

The largest  $L$  satisfying the condition is  $L = 5$ .

# Solution — Exercise 13.5 (2/2)

## Solution

**Step 3 — reject.** BH rejects the 5 smallest hypotheses  $H_{(1)}, \dots, H_{(5)}$  (everything except  $p = 0.600$ ).

**Comparison on identical data.**

Bonferroni: 2 < Holm: 3 < BH: 5 rejections.

Controlling the FDR yields many more discoveries than controlling the FWER — at the price of allowing a small expected fraction ( $\leq q$ ) of them to be false. Note BH is a *step-up* rule: even though  $p_{(4)} = 0.030$  and  $p_{(5)} = 0.040$  individually clear their thresholds here, BH would reject them via the largest passing  $L$  regardless of any earlier borderline case.

*Common mistake:* treating  $q$  like  $\alpha$ . They bound different things:  $\alpha$  caps the chance of *any* false rejection;  $q$  caps only the expected *fraction* of the rejected list that is false.

## Extended Exercise 13.1 — Full workflow on ten p-values [Math]

## Extended exercise (15 min)

Ten hypotheses give the sorted  $p$ -values

$$p = (0.0008, 0.0041, 0.0060, 0.0100, 0.0180, 0.0290, 0.0850, 0.150, 0.240, 0.790),$$

with  $m = 10$ . Control the FWER at  $\alpha = 0.05$  (Bonferroni, Holm) and the FDR at  $q = 0.10$  (Benjamini–Hochberg).

- 1 List the rejections under **Bonferroni**, **Holm** and **BH**.
- 2 Verify the ordering  $R_{\text{Bonf}} \leq R_{\text{Holm}} \leq R_{\text{BH}}$ .
- 3 State the guaranteed FWER bound, and comment on what the FDR guarantee buys.

## Solution — Extended Exercise 13.1 (1/3)

Bonferroni and Holm ( $\text{FWER} \leq 0.05$ )

**Bonferroni:** reject if  $p_j \leq \alpha/m = 0.005$ . Only  $p_{(1)} = 0.0008$  and  $p_{(2)} = 0.0041$  qualify  $\Rightarrow$  **2 rejections**.

**Holm:** compare  $p_{(j)}$  with  $\alpha/(m - j + 1)$ , stopping at the first failure:

| $j$ | $p_{(j)}$ | $\alpha/(m - j + 1)$ |        |
|-----|-----------|----------------------|--------|
| 1   | 0.0008    | $0.05/10 = 0.00500$  | ✓      |
| 2   | 0.0041    | $0.05/9 = 0.00556$   | ✓      |
| 3   | 0.0060    | $0.05/8 = 0.00625$   | ✓      |
| 4   | 0.0100    | $0.05/7 = 0.00714$   | × stop |

Holm rejects ranks 1–3  $\Rightarrow$  **3 rejections**.

## Solution — Extended Exercise 13.1 (2/3)

Benjamini–Hochberg ( $\text{FDR} \leq 0.10$ )

Compare each  $p_{(L)}$  with  $Lq/m = 0.01L$  and take the *largest* passing  $L$ :

| $L$ | $p_{(L)}$ | $Lq/m$ |             |
|-----|-----------|--------|-------------|
| 5   | 0.0180    | 0.050  | ✓           |
| 6   | 0.0290    | 0.060  | ✓ ← largest |
| 7   | 0.0850    | 0.070  | ×           |
| 8   | 0.150     | 0.080  | ×           |

(Ranks 1–4 pass too.) The largest  $L$  with  $p_{(L)} \leq Lq/m$  is  $L = 6$ , so BH rejects the six smallest  $\Rightarrow$  **6 rejections** — including  $p_{(6)} = 0.029$  and ranks 4, 5 that the FWER rules missed.

## Solution — Extended Exercise 13.1 (3/3)

## Comparison and error guarantees

$$\underbrace{2}_{\text{Bonferroni}} \leq \underbrace{3}_{\text{Holm}} \leq \underbrace{6}_{\text{BH}}.$$

**FWER bound.** Bonferroni and Holm both guarantee  $\text{FWER} = \Pr(V \geq 1) \leq \alpha = 0.05$  under *any* dependence: the chance of *even one* false rejection is at most 5%. **What FDR buys.** BH instead guarantees

$\text{FDR} = E[V/R] \leq q m_0/m \leq 0.10$ . Among its 6 discoveries, at most an *expected* 10% ( $\lesssim 1$  here) are false. Tolerating a small false *fraction* rather than *any* false positive is exactly why BH rejects more.

*Common mistake:* reading “ $\text{FDR} \leq 0.10$ ” as “each discovery is 90% likely true”. The FDR is an average over the whole rejected list (and over repetitions), not a per-discovery probability.



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# $p$ -hacking and selection bias

## Don't

- Report only the most-significant subset of tests.
- Test thousands of hypotheses and call the smallest  $p$ -value “significant”.

## Takeaway

Multiple-testing correction must include *every* test you ran, not only the ones you report. Pre-register hypotheses where possible; out-of-sample replication is the gold standard.

## Extended Exercise 13.2 — FWER or FDR? A design decision [Integrative]

### Extended exercise (15 min)

For each study, choose an error criterion and justify it, using the naive expected number of false discoveries  $E[V] = m_0 \alpha$  (all-null worst case  $m_0 = m$ ) and the power consequences.

- 1 **Scenario A — confirmatory phase-III trial:**  $m = 4$  co-primary endpoints; one false “significant” endpoint could license an ineffective drug.
- 2 **Scenario B — exploratory screen:**  $m = 20\,000$  genes tested for follow-up; a shortlist with a few false leads is acceptable.
- 3 Contrast the statistical power the two criteria leave on the table.

# Solution — Extended Exercise 13.2 (1/3)

## Scenario A — confirmatory trial: control FWER

With  $m = 4$  tests at  $\alpha = 0.05$ , naive testing expects

$$E[V] = m_0\alpha = 4(0.05) = 0.20 \text{ false positives,} \quad \text{FWER} = 1 - 0.95^4 = 0.185.$$

A  $\sim 19\%$  chance of *any* false endpoint is unacceptable when the decision is regulatory and irreversible. **Choose FWER control** (Bonferroni threshold  $\alpha/m = 0.0125$ , or Holm). One false positive is the outcome we must not allow, so we pay the power cost gladly — with only  $m = 4$  tests that cost is tiny.

## Solution — Extended Exercise 13.2 (2/3)

### Scenario B — exploratory screen: control FDR

With  $m = 20\,000$  and (worst case)  $m_0 \approx 20\,000$ , naive testing expects

$$E[V] = m_0\alpha = 20\,000(0.05) = 1000 \text{ false discoveries!}$$

Bonferroni would shrink the threshold to  $\alpha/m = 2.5 \times 10^{-6}$ , cutting  $E[V]$  to  $\approx 0.05$  but **destroying power** for realistic effect sizes. **Choose FDR control** (BH at  $q = 0.10$ ): accept that an expected  $\leq 10\%$  of the shortlist is false in exchange for detecting far more true signals — the false leads are removed by follow-up experiments.

## Solution — Extended Exercise 13.2 (3/3)

## The power trade-off

|                 | Confirmatory ( $m = 4$ ) | Exploratory ( $m = 20\,000$ ) |
|-----------------|--------------------------|-------------------------------|
| Criterion       | FWER (Bonferroni/Holm)   | FDR (BH)                      |
| Threshold shape | $\alpha/m$ (fixed, tiny) | adaptive $Lq/m$               |
| Naive $E[V]$    | 0.20                     | 1000                          |
| Priority        | no false positive        | many true findings            |

**Trade-off.** The FWER threshold shrinks like  $\alpha/m$ , so power collapses as  $m$  grows — fine for a handful of endpoints, ruinous for 20 000 genes. FDR asks only that false positives be a small *fraction* of discoveries, so its adaptive threshold keeps power high at scale. **Rule of thumb:** match the criterion to the cost of a single false positive.

*Common mistake:* quoting  $E[V] = m\alpha$  as a fact about your study — it is the all-null worst case ( $m_0 = m$ ). With real effects present the count is smaller, though still ruinous at scale.

# Outline

- 1 The Problem
- 2 FWER
- 3 FDR
- 4 Practice
- 5 Python Lab**
- 6 Summary

# Python lab — run it live in the notebook

## Companion notebook: `chapter_13_lab.ipynb`

The code in this section is meant to be **demonstrated live**. Open the companion Jupyter notebook and run it cell by cell:

- simulating  $m = 1000$  tests with a fraction of true effects;
- comparing Bonferroni, Holm and Benjamini–Hochberg;
- visualising the BH threshold line and a permutation test.

Data loads via the ISLP package or the bundled CSV files; the notebook ends with hands-on exercises.



# Bonferroni, Holm, BH in statsmodels

```
from statsmodels.stats.multitest import multipletests # corrections
import numpy as np # arrays

p = np.array([1e-6, 0.01, 0.04, 0.05, 0.30, 0.45, 0.80]) # raw p-values

for method in ["bonferroni", "holm", "fdr_bh"]: # FWER, FWER, FDR control
    rej, p_corr, *_ = multipletests(p, alpha=0.05, method=method) # adjust
    print(f"{method:12s}", rej) # rejection flags (bool)
    print(f"{'p_adj':12s}", np.round(p_corr, 4)) # adjusted p-values
```

## Extended Exercise 13.3 — Simulating FDR and power [Python]

### Extended exercise (15 min)

Simulate  $m = 1000$  one-sided z-tests of which  $m_1 = 100$  have a true effect (mean shift  $\mu = 3$ ); the other 900 are null. Over many repeats, apply Benjamini–Hochberg at  $q = 0.10$  and Bonferroni at  $\alpha = 0.05$ , and estimate the *realised* FDR ( $V/R$ ) and power ( $S/m_1$ ) of each.

- 1 Write the simulation and run  $\sim 2000$  repeats.
- 2 Report the realised FDR and power for BH and for Bonferroni.
- 3 Check BH against its theoretical bound  $q m_0/m$ , and comment on the power gap.

## Solution — Extended Exercise 13.3 (1/3)

## Code: setup and the BH rule

```
import numpy as np # arrays and random numbers
from scipy import stats # normal cdf for p-values
rng = np.random.default_rng(0) # seeded generator: reproducible
m, m1, mu = 1000, 100, 3.0 # 100 true effects, mean shift mu
q, alpha = 0.10, 0.05 # FDR target q, FWER target alpha
truth = np.arange(m) < m1 # first m1 tests are non-null

def bh_reject(p, q): # Benjamini--Hochberg step-up
    order = np.argsort(p); ps = p[order] # sort p-values ascending
    thr = (np.arange(1, m + 1) / m) * q # BH line: (j/m) * q
    hit = np.where(ps <= thr)[0] # ranks below the line
    rej = np.zeros(m, bool) # start: reject nothing
    if len(hit): rej[order[:hit.max() + 1]] = True # up to largest hit
    return rej # boolean rejection mask
```

## Solution — Extended Exercise 13.3 (2/3)

## Code: simulate FDR and power

```
def simulate(method, reps=2000): # average FDR / power over reps
    fdr = pw = 0.0 # running totals
    for _ in range(reps):
        z = rng.normal(0, 1, m); z[:m1] = rng.normal(mu, 1, m1) # effects: mean mu
        p = 1 - stats.norm.cdf(z) # one-sided p-values
        rej = bh_reject(p, q) if method == "bh" else (p <= alpha / m) # BH or Bonf.
        V = (rej & ~truth).sum(); S = (rej & truth).sum(); R = rej.sum() # V, S, R
        fdr += V / max(R, 1); pw += S / m1 # realised FDP and power
    return fdr / reps, pw / reps # averages over the repeats

for meth in ["bh", "bonferroni"]: # compare both procedures
    f, pw = simulate(meth) # run the experiment
    print("%-11s FDR=%.3f power=%.3f" % (meth, f, pw)) # summary line
```

# Solution — Extended Exercise 13.3 (3/3)

## Expected numbers and interpretation

### Typical printout.

```
bh    FDR=0.090  power=0.723
bonferroni  FDR=0.002  power=0.186
```

- **BH** holds the realised FDR at 0.090 — essentially its theoretical ceiling  $q m_0/m = 0.10 \times 900/1000 = 0.09$  — while recovering about **72%** of the true effects.
- **Bonferroni** is far stricter: realised FDR  $\approx 0.002$ , but power collapses to about **19%** — it misses four-fifths of the signals.
- **Lesson:** at  $m = 1000$ , FDR control detects roughly *four times* as many true effects as FWER control, for a small, bounded false fraction.

*Common mistake:* expecting the realised FDR to hit  $q = 0.10$  exactly. The guarantee is  $q m_0/m = 0.09$  here — BH adapts to the unknown share of true nulls, and any single run fluctuates around it.

# Outline

- 1 The Problem
- 2 FWER
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# Chapter 13 in one slide

## What we accomplished

Honest inference when many hypotheses are tested at once.

- Set up the multiple-testing problem.
- Family-wise error rate (FWER): Bonferroni and Holm.
- False discovery rate (FDR): Benjamini–Hochberg.
- Resampling-based  $p$ -values and Storey's  $q$ -values.
- Pitfalls of post-selection inference and  $p$ -hacking.

# Key formulas at a glance

**FWER**  $\text{FWER} = \Pr(V \geq 1) = 1 - (1 - \alpha)^m$  (independent tests).

**Bonferroni** reject  $H_{0j}$  if  $p_j \leq \alpha/m$  (controls FWER).

**Holm** reject while  $p_{(j)} \leq \frac{\alpha}{m-j+1}$ , stepping down from the smallest.

**FDR**  $\text{FDR} = E\left[\frac{V}{\max(R, 1)}\right]$  (expected false-discovery proportion).

**Benjamini–Hochberg** reject the  $L$  smallest, where  $L = \max\{j : p_{(j)} \leq \frac{j}{m}q\}$ .

**Power**  $\text{Power} = E[S]/m_1$ .



# Vocabulary checklist

| Term                          | Meaning                                       |
|-------------------------------|-----------------------------------------------|
| Type I error                  | Reject $H_0$ when true                        |
| Type II error                 | Fail to reject when false                     |
| $\text{FWER} = \Pr(V \geq 1)$ | Family-wise error rate                        |
| $\text{FDR} = E[V/R]$         | False discovery rate                          |
| Bonferroni / Holm             | FWER controls                                 |
| BH procedure                  | Step-up FDR control                           |
| Storey's $q$ -value           | FDR-calibrated $p$ -value                     |
| Permutation test              | Null distribution by relabelling              |
| Knockoffs                     | Modern FDR control with covariate engineering |

# Decision rules of thumb

- Always specify the family of tests *before* looking at the data.
- Confirmatory / regulated  $\Rightarrow$  FWER (Bonferroni or Holm).
- Exploratory / discovery  $\Rightarrow$  BH on FDR.
- Report both raw and adjusted  $p$ -values.
- Replicate findings on an independent sample whenever possible.
- For valid post-selection inference: sample-splitting, selective inference, knockoffs.

# Common pitfalls

## Don't

- 1 Test thousands of hypotheses and report only the “significant” ones.
- 2 Use BH when tests are strongly negatively dependent.
- 3 Confuse FWER and FDR — they answer different questions.
- 4 Apply post-hoc multiple-testing fixes after looking at the data.
- 5 Use uncorrected  $p$ -values from stepwise / lasso.
- 6 Treat a  $p$ -value as a probability that  $H_0$  is true.

# Course-wide summary

## The conceptual arc

- 1 Vocabulary and bias-variance (Ch. 1–2).
- 2 Linear baselines and how to evaluate them (Ch. 3–5).
- 3 Regularisation and dimension reduction (Ch. 6).
- 4 Adding flexibility: splines, GAMs, trees, SVMs, deep nets (Ch. 7–10).
- 5 Special responses: censored times, no response (Ch. 11–12).
- 6 Honest inference at scale (Ch. 13).

# Closing thoughts

## The four habits of an effective statistical learner

- ① Always plot the data.
- ② Always validate on data you didn't train on.
- ③ Always interpret what the model says about the world.
- ④ Always check assumptions and limitations before claiming results.

Thank you for taking this course. Good luck applying statistical learning to your own data!

## Appendix — what is in here, and why

Nothing in the appendix is needed to follow the chapter: each slide is a formal derivation, a longer worked exercise, or a side topic. Read it when you want the full story, or when a later chapter sends you back here.

### Contents of the appendix

- **The four outcomes, drawn** — a diagram of the same  $2 \times 2$  table the main deck already gives as a table.
- **Resampling-based inference** — permutation  $p$ -values, the resampling estimate of the FDR, and what the histogram of many  $p$ -values looks like. Use it when the null distribution is not available in closed form.
- **Post-selection inference** — why a  $p$ -value computed after looking at the data is not a  $p$ -value. Important in research practice, beyond the exam.

# The four outcomes of a test decision

|             | Not rejected                | Rejected                   |       |
|-------------|-----------------------------|----------------------------|-------|
| $H_0$ true  | $U$<br>true negatives       | $V$<br>Type I (false pos.) | $m_0$ |
| $H_0$ false | $T$<br>Type II (false neg.) | $S$<br>true positives      | $m_1$ |
|             | $m - R$                     | $R = V + S$                |       |

## Takeaway

**Type I errors** are the red cell  $V$ . The  $\text{FWER} = \Pr(V \geq 1)$  guards against *any* false positive; the  $\text{FDR} = E[V/R]$  controls their *expected fraction* among the  $R$  rejections.

## Resampling-based $p$ -values

When the test statistic's null distribution is hard to derive:

### How to read this

- **Permutation**: shuffle the response many times to get a null distribution.
- **Bootstrap (under  $H_0$ )**: resample under a constraint that enforces the null.
- Empirical  $p$ -values are then thresholded by Bonferroni / Holm / BH as before.

### Takeaway

Empirical  $p$ -value =  $\frac{1 + \#\{\text{resampled statistics} \geq \text{observed}\}}{1 + B}$  over  $B$  resamples — no parametric null needed. The “+1” keeps it strictly positive.



# Resampling-based FDR

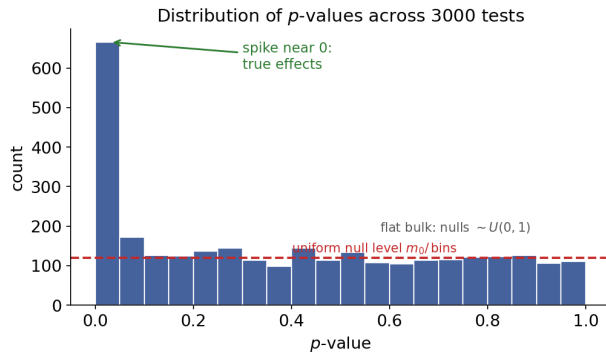
## Storey's $q$ -value

Estimate the proportion of true nulls  $\hat{\pi}_0$  from the  $p$ -value distribution. Calibrate FDR more accurately than BH. Especially relevant when many nulls are false.

## How $\hat{\pi}_0$ is read off

True nulls give  $p$ -values that are  $\sim U(0, 1)$ , so the histogram is *flat* away from 0; the height of that flat part estimates  $\hat{\pi}_0$ . BH implicitly assumes  $\hat{\pi}_0 = 1$ ; plugging in the smaller true value makes FDR control less conservative and recovers more discoveries.

# The distribution of $p$ -values under many tests



## Takeaway

True nulls give  $p$ -values that are  $\sim U(0, 1)$  (the flat bulk), while true effects pile up near 0. Storey's method reads  $\hat{\pi}_0$  off the height of that flat part to calibrate the FDR.

# Post-selection inference

After variable selection (lasso, stepwise) standard  $p$ -values are invalid.

## Takeaway

The bias: the *same data* both *chose* the hypothesis and *tested* it, so the reported  $p$ -values are optimistically small. Valid inference must account for the selection step.

## Modern methods

- Sample splitting / data-carving.
- Selective inference (Lee, Sun, Sun, Taylor).
- Knockoffs (Barber, Candès).